सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/29

Date: 21-05-2024

As per distribution list

Subject: 29th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 29th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 17th May, 2024 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

काशिश 21/05/24

(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 29th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 17.05.2024

The 29th Consultation Meeting for Evolving Transmission Schemes in the Northern Region (CMETS-NR) was held through VC on 17.05.24. List of participants is enclosed at **Annexure-A.**

It was informed that the Minutes of the 28th CMETS-NR meeting held on 27/03/2024 were issued vide letter dated 08/04/2024. No comments were received on the minutes, accordingly, minutes were confirmed as circulated.

A. Network Expansion Scheme

A1. Augmentation with 400/220 kV 1x500 MVA(9th) ICT at Bikaner-II to meet N-1 compliance

It was stated that 400/220 kV Bikaner-II PS is an existing RE pooling station with transformation capacity of 1000 MVA(2x500MVA). Further, 6 no. of 500 MVA 400/220kV, ICTs are under implementation with the schedule progressively from Jun'24 upto Jan'25. Therefore, considering above total transformation capacity at 400/220 kV Bikaner-II PS shall be 4000MVA (8x500MVA) by Jan'25. Considering the total connectivity of 3785 MW granted at 220 kV level at Bikaner-II PS, augmentation of 1x500 MVA, 400/220kV (9th) ICT at Bikaner-II PS shall be required to meet N-1 compliance.

Grid-India (NRLDC) enquired about bus splitting arrangement at 220kV level of Bikaner-II PS. CTUIL stated that there is bus sectionalizer arrangement (normally closed) as per present practise. Grid-India stated that in some of RE pooling stations there are 2 or 3 different 220kV sections with n-1 compliance (400/220kV ICT) in each section. CTU stated that in some pooling stations i.e. Fatehgarh-II PS, Bhadla-II PS where 220kV bus sections are geographically isolated (way apart from each other), accordingly there is no sectionalizer arrangement feasible in two sections and n-1 compliance of 400/220kV ICTs to be ensured Iin both the sections as per planning criteria. CTU also stated that different configuration of 220kV section arrangement i.e. 220kV isolated sections in two different directions or 220kV common bus with sectionalizer arrangement (normally closed) is only decided by the TSP based on GA and layout of substation. Based on above n-1 compliance (400/220kV ICT) requirement will be identified. At Bikaner-II PS there is common 220kV bus with bus sectionalizer arrangement which would be normally in closed position, therefore N-1 compliance is being considered on common 220kV bus. Grid-India stated that some common philosophy should be adopted for Grid operation.

No other comments were received on ICT augmentation scheme at Bikaner-II PS. In view of the above, following ICT augmentation scheme is agreed at Bikaner-II PS:

- Augmentation with 400/220 kV, 1x500 MVA (9th) ICTs at Bikaner-II PS along with associated transformer bays

A2. Augmentation with 400/220 kV 3x500 MVA(6th to 8th) ICTs at Fatehgarh-IV PS(Sec-II)

It was deliberated that at present, total connectivity granted at Fatehgarh-IV PS(Sec-II) is 4980 MW(4880+100 MW on hold due to reallocation matter currently subjudice). Out of this, 3480 MW(3380+100 on hold) is granted at 220 kV and 1500 MW is granted at 400 kV level. Establishment of Fatehgarh-IV PS(Sec-II) and associated transmission system is currently under bidding as part of “Transmission System for evacuation of power from Rajasthan REZ Ph-IV (Part-2:5.5 GW) (Jaisalmer/ Barmer Complex)” and expected to be awarded in May’24.

The above scheme involves establishment of 765/400/220 kV Fatehgarh-IV PS(Sec-II) along with transformation capacity of 4x1500 MVA 765/400 kV ICTs & 5x500 MVA 400/220 kV ICTs. Considering the connectivity of 3480 MW granted at 220 kV level of Fatehgarh-IV PS(Sec-II), augmentation of 3x500 MVA ICTs are required to be taken up matching with the above scheme for evacuation of power as well as to meet N-1 compliance.

Grid-India enquired about RE generation being interconnected at Fatehgarh-IV PS(Sec-I). CTU stated that at Fatehgarh-IV PS (Sec-I), about 2025MW connectivity is granted and both the sections i.e. section-I & II shall be electrically isolated through sectionalizer (normally open) with compliance will be separate for both the sections. Accordingly, N-1 of 400/220kV ICTs is being ensured in both the sections. CEA agreed for the proposal.

Accordingly, the following ICT augmentation scheme is agreed at Fatehgarh-IV PS(Sec-II) PS in ISTS :

- Augmentation with 400/220 kV, 3x500 MVA (6th ,7th & 8th) ICTs at Fatehgarh-IV PS(Sec-II) along with associated transformer bays

A3. Augmentation with 400/220 kV 1x500 MVA(3rd & 4th) ICT at Barmer-I PS

At present total connectivity granted at Barmer-I PS is 3050 MW at 220 kV. Out of this 3050 MW, 1350 MW connectivity is granted with Transmission System for evacuation of power from Rajasthan REZ Ph-IV (Part-2:5.5 GW) (Jaisalmer/ Barmer Complex) scheme. The connectivity beyond 1350 MW(upto 4000 MW) is being granted with Rajasthan REZ Ph-IV (Part-4:3.5 GW) scheme which was recently approved in the 19th NCT meeting held on 29.04.2024.

Establishment of 765/400/220kV Barmer-I PS (765/400KV: 3x1500MVA, 400/220KV: 2x500MVA) and associated system is currently under bidding as part of “Transmission System for evacuation of power from Rajasthan REZ Ph-IV (Part-2:5.5 GW) (Jaisalmer/Barmer Complex)” and is expected to be awarded in May’24. Therefore, considering the connectivity of 1350 MW granted at 220 kV level of Barmer-I PS with Ph-IV Part-2 scheme, augmentation of 1x500 MVA ICT (3rd) is required to be taken up matching with the above scheme(Rajasthan REZ Ph-IV (Part-2:5.5 GW)) for reliable evacuation of power.

CEA stated that with 1350MW quantum at Barmer-I PS, N-1 compliance also needs to be taken care and therefore 2 nos. of 400/220kV ICTs (3rd & 4th) shall be required for evacuation as well as to meet N-1 requirement. CTU stated that considering 4000MW RE quantum (Potential) at 220kV level of Barmer-I PS, total 9 no's, of ICTs are required for evacuation as well as to meet N-1 compliance. Out of 9 nos. of ICTs, 2nos. of ICTs (1st & 2nd) were approved as part of Rajasthan REZ Ph-IV (Part-2:5.5 GW) scheme and 5x500MVA ICTs (5th to 9th) is also approved in recent NCT meeting as part of Rajasthan REZ Ph-IV (Part-4:3.5 GW) scheme and with proposed augmentation (3rd ICT), total 8 nos. ICTs is being considered whereas 9th ICT can be taken at later stage to meet N-1 compliance requirement. CEA stated that there is some time gap (5-6 months) in commissioning of two schemes i.e. REZ Ph-IV (Part-2:5.5 GW) & REZ Ph-IV (Part-4:3.5 GW) and for such period 400/220kV ICTs will be n-1 non-compliant. CEA stated that 2x500MVA (3rd & 4th) may be considered. SECI enquired whether GNA Operationalization of RE applicants at Barmer-I PS may also be affected in absence of 4th ICT required to meet N-1 compliance. CTU stated that GNA Operationalization of RE applicants will not be dependent on ICT required to meet N-1 compliance, however considering discussions in meeting, 400/220 kV, 2x500 MVA (3rd & 4th) ICTs at Barmer-I PS is considered to meet power evacuation as well as transformer n-1 requirement.

Accordingly, the following ICT augmentation scheme is agreed at Barmer-I PS in ISTS:

- Augmentation with 400/220 kV, 2x500 MVA (3rd & 4th) ICTs at Barmer-I PS along with associated transformer bays

A4. Augmentation with 500 MVA (4th) ICT at 400/220 kV Bhiwadi (Hybrid) substation

It was stated that in 28th CMETS-NR meeting held on 27.03.24, agenda for augmentation of 400/220kV ICT at Bhiwadi Substation was deliberated. In the meeting it was discussed that from the loading pattern of Bhiwadi ICTs (3x315 MVA) , it is observed that cumulative maximum loading on 400/220 kV ICTs is about 820 MW and loading is higher (>600MW) in Jun-Sep'23 & Jan'24-Feb'24 for sufficient duration of time. From the analysis, it emerged that loading is breaching N-1 limit for peak Loading of ICTs (800 MW). Further in CTU planning file, ICT loadings are higher (2x310 MW in case of N-1 Contingency) and may breach the N-1 limit. The loading pattern of Bhiwadi ICTs for past one year is as below (Fig 1):

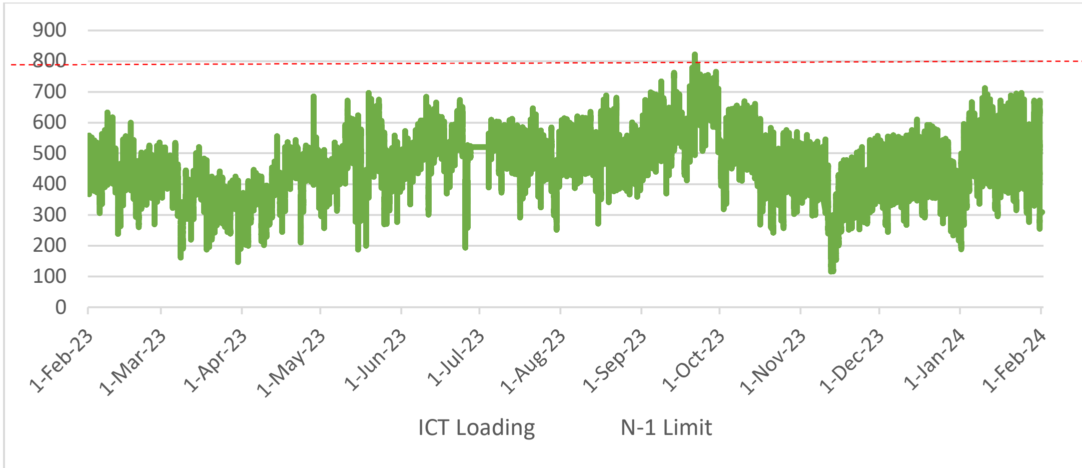


Fig 1: Loading of Bhiwadi ICTs of Past one year (Source: Grid-India)

POWERGRID vide mail dated 23.02.24 confirmed space availability for Augmentation of 400/220kV transformer (4^{th},1x500MVA) along with transformer bays at 400/220kV Bhiwadi (PG) S/s. However, 220kV Side of ICT is required to be routed through cable.

In the meeting Grid-India stated that at Bhiwadi transformer loading are close to N-1 limit and in view of that requirements may be assessed based on future load growth in Bhiwadi complex considering planned interconnections by RVPN/HVPN in future. RVPN stated that at present for Rajasthan, 4 nos. of 220kV feeders are interconnected at Bhiwadi substation and no new interconnection is planned for drawl in future. RVPN also stated that a new 400/220kV substation near Bhiwadi is recently approved by RVPN board to cater future drawl requirement at Bhiwadi complex in view of anticipated load growth. In reply to implementation timeframe of proposed new substation in Bhiwadi, RVPN stated that implementation may take minimum 3 years from now.

POWERGRID stated that loading at Bhiwadi substation are critical and N-1 noncompliant in peak load condition. Further, 2 nos. of 400/220kV ICTs (make 2003 year) already completed 21 years of life and gases are observed to be high due to loading of ICTs. Considering above ICT augmentation is required at Bhiwadi substation. CTU stated that ICT loadings are close to N-1 limit, however HVPN/RVPN need to concur on requirements of ICT augmentation. CTU stated that decision of ICT augmentation may be expedited as Implementation of 400/220kV ICT may take 18-21 months from award and ICT may be available by mid of 2026 only.

Representative from HVPN were not present in the last CMETS-NR meeting . RVPN stated that they will re-assess the load pattern at 400/220kV Bhiwadi substation along with future load growth requirement in complex and therefore requested to take up agenda in next meeting along with

HVPN views for their drawl requirement. Considering above, it was decided that agenda for augmentation at 400/220kV Bhiwadi S/s will be taken up in this CMETS-NR meeting.

CTU enquired RVPN and HVPN to update on the requirement of Augmentation of 400/220kV transformer (4^{th},1x500MVA) at Bhiwadi S/s based on their assessment.

RVPN stated that power flow over all the existing drawl feeders from Bhiwadi (PG) is already saturated and to meet the future load of Bhiwadi area, other scheme is being planned. RVPN also stated that 220kV Bhiwadi(PG)-Neemrana (RVPN) is being LILOed at Neemrana (PG) and with above LILO part load (~50-70MW) will be shifted from Bhiwadi(PG) to Neemrana(PG) and RVPN may not drawl additional power from existing feeders of Bhiwadi(PG) S/s.

HVPN stated that at present no additional drawl requirement is envisaged in future from Bhiwadi (PG) S/s. However for N-1 compliance, HVPN concurred for Augmentation of 400/220 kV, 500 MVA (4th) ICT at Bhiwadi S/s and same was informed vide HVPN letter dated 17.05.24.. HVPN also stated that they are planning some ICT augmentation at 220kV intra state network, however it will not impact 400/220kV ICT loading much.

Grid-India suggested that as N-1 noncompliance occurs for short duration of time, replacement of 315MVA ICT with 500MVA ICT may be considered in place of augmentation with 4^{th} ICT at Bhiwadi S/s.

POWERGRID stated that even with load diversion (~50-70MW), ICT loadings are close to N-1 limits and considering type of load (critical) as well as with increasing gases trend in existing ICTs, ICT augmentation may be considered. POWERGRID stated that 2nos. of ICTs at Bhiwadi S/s requires continuous inspection & degassing due to high loading of ICTs and they are not able to get shutdown of ICTs from RVPN side due to continuous load demand in Bhiwadi complex and with long outage of ICTs, load will also be impacted.

RVPN stated that in long outage condition of ICTs, there will be severe issues in serving the critical load of Bhiwadi complex which is mainly industrial load and replacement of ICT may also require shutdown. RVPN stated that in view of issues in existing 400/220kV ICTs, frequent shutdown requirement and possible long outages of above ICTs which will impact critical industrial loads in Bhiwadi area , augmentation with 4^{th} ICT may be considered at Bhiwadi substation . HVPN also concurred on the same.

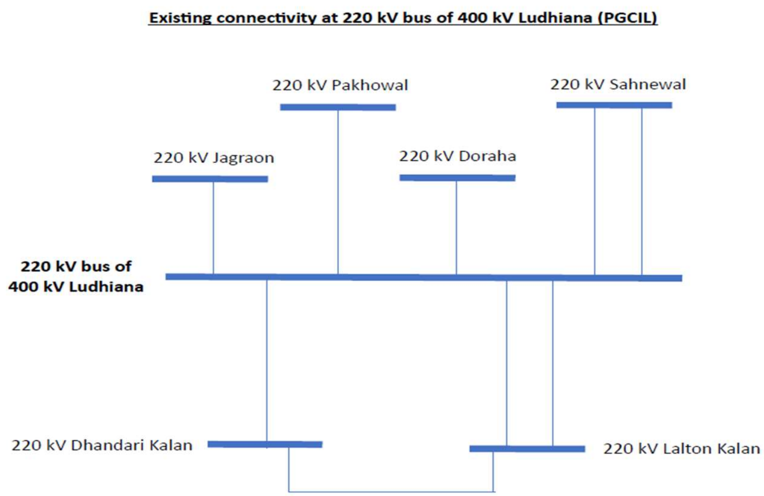
Grid-India stated that as RVPN and HVPN concurred for ICT augmentation requirement to meet N-1 compliance, same is agreeable to them instead of replacement of ICT.

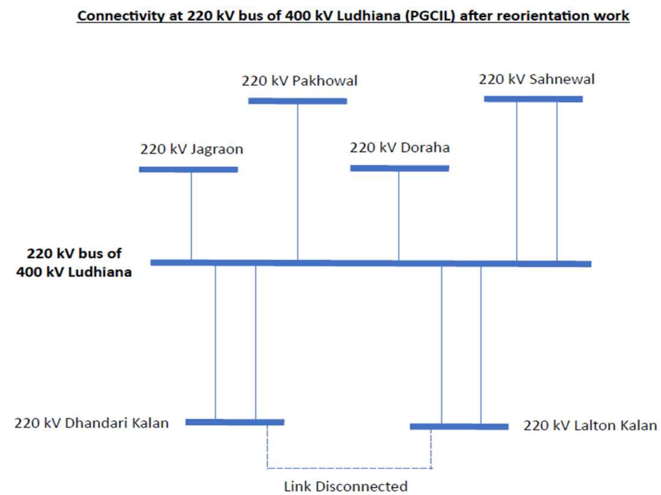
In view of above deliberations, following ICT augmentation scheme is agreed in ISTS:

- Augmentation with 400/220 kV, 500 MVA (4th) ICT at Bhiwadi (PG) S/s along with associated transformer bays
- Cable requirement on 220kV Side of ICT

A5. Bypassing of 220 kV Dhandari Kalan (PSTCL) line from 220 kV Lalton Kalan (PSTCL) to 400 kV Ludhiana (PG)

It was deliberated that PSTCL vide their letter dated 07/05/2024 had informed that reorientation of the existing 220 kV Dhandari Kalan (PSTCL) - Lalton Kalan (PSTCL) S/c line from Lalton Kalan to 400/220 kV Ludhiana (PG) substation was planned by PSTCL for improving the reliability & reducing fault level in the complex. The scheme is shown as below:





It was informed that the scheme was also approved in the 57th NRPC meeting held on 31/08/2022. However, the subject line was not concurred in any CMETS/Standing Committee Meetings of Northern Region.

Now, this 220 kV Dhandhari Kalan – Ludhiana (PG) line has been implemented and ready for charging and with reference to IEGC 2023 (Indian Electricity Grid Code), for any new ISTS transmission connectivity, a connectivity agreement is required to be signed between STU, CTU and transmission licensee before the physical connection to ISTS grid. Accordingly, PSTCL had submitted Technical Connection data of Dhandhari Kalan to Ludhiana (PG) 400/220 kV Substation 220 kV line to CTU for processing of signing of Connectivity Agreement.

However, as stated above, the scheme was not approved in any Technical/NR-CMETS meeting and looking into the urgency of the matter (Line is ready for charging), the agenda for concurrence of the subject line was circulated through email on 08/05/2024 for any observations/comments. It was also mentioned that in case of non-receipt of any comments from stakeholders, the agenda shall be noted in the upcoming NR-CMETS meeting. Further, no comments/observations were received on the re-orientation of the said 220 kV line till date. Based on this, Connection Details have been issued on 13/05/2024 and Connectivity Agreement was signed on 17/05/2024.

In view of above, the matter of the reorientation of the existing 220 kV Dhandari Kalan (PSTCL) - Lalton Kalan (PSTCL) S/c line from Lalton Kalan to 400/220 kV Ludhiana (PG) substation was noted.

A6. Replacement of 400 kV Fatehgarh-II – Bhadla-III D/c line from the Transmission system for Connectivity/LTA/GNA

It was deliberated that as part of Transmission system for REZ in Rajasthan Phase-III (20 GW) scheme, 400 kV Fatehgarh-II – Bhadla-III D/c line was approved in the 5th NCT meeting held on 25.08.2021 & 02.09.2021 for implementation in TBCB mode as part of Phase-III Part B1 package. Subsequently, MoP vide gazette dated 03.12.2021 appointed PFCCL as the BPC of the Phase-III Part B1 package.

However, Hon'ble Supreme court constituted GIB committee vide letter dated 31.01.2023 informed that laying of above line is very risky for the GIB, as two powerline induced mortalities had been reported recently from that area and therefore asked to explore alternatives in this regard. CTU vide letter dated 23/02/23 informed that any other alternative shall not help in avoiding GIB prioritized area in Rajasthan, therefore The GIB committee was requested to consider ratification of above line with rerouting through alternative route to run along existing/under construction line to the extent possible or any other route in Degray Oran area. However, no clearance was received from the GIB Committee for above line.

Subsequently, a meeting was held on 01.05.2023 under the chairmanship of Secretary, MoP, to review the progress of under construction/ under bidding/ planned Transmission Projects for evacuation of Renewable Energy (RE) projects, wherein it was highlighted that the GIB clearance for Fatehgarh-II PS – Bhadla-III PS 400 kV D/c line has not been received due to which the bidding process for the transmission scheme is getting delayed. Considering that, Secretary, MoP, directed that the process of delinking of 400 kV Fatehgarh-II- Bhadla-III D/c line from Phase-III Part-B1 may be carried out at the earliest and a separate package may be formed comprising 400 kV Fatehgarh-II- Bhadla III D/c line.

Considering above, delinking of 400 kV Fatehgarh-II- Bhadla-III D/c line from Phase-III Part B1 package was discussed and agreed in the 14th NCT meeting held on 09.06.2023. Subsequently, Ph-III Part B1 (without 400 kV Fatehgarh-II-Bhadla-III D/c line) was awarded by PFCCL and currently under implementation.

400kV Fatehgarh-II- Bhadla-III D/c line was part of transmission system for LTA of some of the earlier grantees as this line was required to provide necessary anchoring support for Bhadla-III PS at 400 kV level. However, as the line was not ratified by the GIB committee, implementation of the line was deferred.

Further, as part of Rajasthan Ph-III Part A3 scheme, 400 kV Fatehgarh-III – Bhadla-III D/c line along with associated reactive compensation is already awarded and currently under implementation (Sch. Feb'25). The 400 kV Fatehgarh-III PS-Bhadla-III PS shall provide necessary anchoring support at 400 kV level of Bhadla-III PS.

In view of the above, it is proposed to replace 400 kV Fatehgarh-II – Bhadla-III D/c line from the transmission system for LTA (now GNA) with 400 kV Fatehgarh-III – Bhadla-III D/c line(Ph-III Part A3) to the grantees . It was also stated that with proposed replacement, there is no impact on start date of connectivity as per the SCOD of the transmission schemes due to above change in the system to Connectivity/LTA/GNA grantees.

No comments received on above proposal and therefore replacement of 400 kV Fatehgarh-II – Bhadla-III D/c line from the transmission system for LTA (now GNA) with 400 kV Fatehgarh-III – Bhadla-III D/c line (Ph-III Part A3) to the grantees was agreed.

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Annexure-A

List of Participants of 29th Consultation meeting for Evolving Transmission Schemes in NR held on 17.05.2024

CEA

Shri Nitin Deswal	Deputy Director
Shri Kanhaiya Singh Kushwaha	Assistant Director

NRPC

Shri D.K.Meena	SE(O)

Grid India

Shri Priyam Jain	Ch. Manager
Shri Gaurav Malviya	Manager
Shri Gaurab Dash	Dy. Manager

SECI

Shri R.K.Agarwal	Consultant

CTUIL

Shri V Thiagarajan	Sr. GM
Shri Kashish Bhambhani	GM
Shri Sandeep Kumawat	DGM
Smt. Ankita Singh	Ch. Manager
Shri R Narendra Sathvik	Ch. Manager

Shri Madhusudan Meena	Engineer
PGCIL	
Shri Rohit Kumar Jain	DGM
HVPNL	
Shri Sanjay Verma	SE Planning
Shri Deepak Sarit	XEN/SS
Smt. Palak Sinha	AE/SS
RVPNL	
Dr. Om Prakash Mahela	XEN(PP&D)
PSTCL	
Shri Nitin Kumar	Sr. XEN
JKPTCL	
Shri Kamal Kishore Thappa	SE
Shri Ajaj Ahmed	SE
CED (Chandigarh)	
Shri Uma Kant Patel	EE
UPPTCL	
Shri. Satendra Kumar	SE